

Advanced Optimization Concepts

Saurabh,

Adjunct Faculty, WILP, BITS-Pilani

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1. The Challenge of Overfitting

In machine learning, our ultimate goal is not just to build a model that is accurate on the data it was trained on. The true test of a model is its ability to make correct predictions on new, unseen data. This ability is called **generalization**.

Overfitting is the single biggest threat to generalization. It occurs when a model learns the training data *too well*. Instead of capturing the underlying trend, the model starts to memorize the noise and random fluctuations in the training set.

Analogy: A Student Cramming for an Exam

- **A Good Student (Good Model):** Learns the underlying concepts and principles of a subject. They understand the "why" behind the formulas. Because they have a deep understanding, they can answer questions they have never seen before. This is generalization.
- **A Student Who Crams (Overfit Model):** Memorizes the exact answers to the practice questions. They get 100% on the practice test, but when the real exam has slightly different questions on the same topics, they fail. They never learned the concepts, only the specific examples.

Just like the cramming student, an overfit model is brittle and not truly intelligent. It gives a false sense of high performance.

Numerical Example of Overfitting

Suppose we want to predict a person's height (y) based on their weight (x_1) and three irrelevant features. Let's assume the true relationship is simply **height $\approx$ weight**.

Consider this (simplified) data:

Weight (x_1)	Fav. Num (x_2)	Street Len (x_3)	Shoe Size (x_4)	Height (y)
61	2	3	8	60
40	0	4	7	40
68	0	10	9	70

We want to find weights for the model $\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_0$.

- **A Good, Generalizing Solution:** $w_1 = 1, w_2 = 0, w_3 = 0, w_4 = 0, w_0 = 0$. This model captures the real relationship ($\hat{y} \approx x_1$). It won't have zero error on the training data, but it will perform well on new people because it has learned the correct pattern.
- **An Overfit Solution:** It's possible to find a complex combination of weights, like $w_1 = 0, w_2 = 7, w_3 = 5, w_4 = 0, w_0 = 20$, that produces **zero error** on this specific training set. However, this model is useless. It thinks height depends on irrelevant features. When it sees a new person, its prediction will be wildly inaccurate.

Key Takeaway: The goal is not to achieve zero training error. The true goal is to build a model that generalizes well to unseen data.

2. Feature Processing: The Importance of Scale

Gradient Descent can struggle when input features are on vastly different scales. Imagine one feature ranges from 0 to 1, while another ranges from 0 to 1,000,000. The loss function becomes much more sensitive to changes in the larger-scale feature.

Analogy: The Lost Tourist

Imagine a tourist in a city where the map is distorted. One block north-south covers 1 kilometer, but one block east-west covers only 10 meters. The tourist wants to go to a location that is 5 blocks north and 5 blocks east. If they take equal "steps" in each direction, they will massively overshoot their target in the north-south direction while barely moving east-west. They will have to make many tiny corrections, zigzagging their way to the destination. This is exactly what happens to Gradient Descent with unscaled features. It wastes time oscillating instead of moving directly toward the solution.

When features have different scales, the loss function becomes a steep, elliptical bowl. Gradient Descent will oscillate back and forth across the steep walls instead of smoothly moving to the bottom.



Scaled Features (Circular Bowl) **Unscaled Features (Elliptical Bowl)**

To fix this, we scale our features. Two common methods are:

1. **Mean Centering:** Subtract the mean of the feature from each data point. This centers the data around zero. Formula: $x' = x - \mu$.
2. **Standardization (Normalization):** After mean centering, divide by the standard deviation. This gives the feature a mean of 0 and a standard deviation of 1. Formula: $x_{scaled} = \frac{x - \mu}{\sigma}$.

Numerical Example of Feature Scaling

Consider a feature ‘Age’ with values $\{22, 45, 60, 80\}$.

1. **Calculate Mean (μ):** $\mu = \frac{22+45+60+80}{4} = 51.75$.
2. **Calculate Standard Deviation (σ):** $\sigma \approx 22.2$.
3. **Scale each point:**
 - 22 becomes $\frac{22-51.75}{22.2} \approx -1.34$
 - 45 becomes $\frac{45-51.75}{22.2} \approx -0.30$
 - 60 becomes $\frac{60-51.75}{22.2} \approx 0.37$
 - 80 becomes $\frac{80-51.75}{22.2} \approx 1.27$

By doing this for all features, we make the loss function more like a circular bowl, allowing Gradient Descent to converge much faster.

3. Challenges in Gradient-Based Optimization

The path to the minimum is not always easy. Here are common obstacles:

- **Local Optima:** A point that is a minimum within its local neighborhood, but not the true, global minimum. Our foggy mountain climber might descend into a small crater and stop, thinking they have reached the bottom of the entire range.
- **Flat Regions (Plateaus):** Large areas where the gradient is close to zero. Since the step size depends on the gradient’s magnitude, the algorithm will slow to a crawl, making convergence painfully slow.
- **Different Curvatures (Ravines):** As seen with feature scaling, if the function is much steeper in one direction than another, the algorithm will oscillate inefficiently.

4. Introduction to Constrained Optimization

So far, we have assumed we can search anywhere for the minimum. But what if there are rules?

Constrained Optimization deals with finding the minimum or maximum of an objective function subject to one or more constraints.

Analogy: The Farmer's Fence

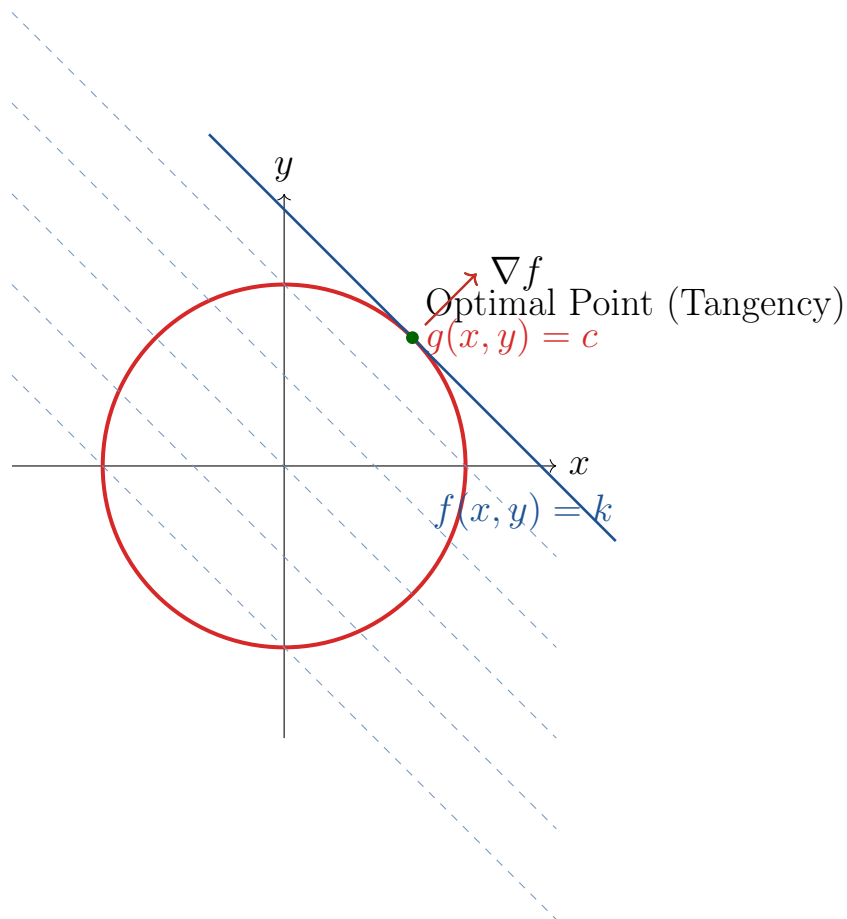
A farmer wants to build a rectangular pen for their sheep. They have exactly **100 meters of fencing**. They want to build the pen with the **largest possible area**.

- **Objective Function:** Maximize Area $A = L \times W$.
- **Constraint:** The perimeter must be exactly 100 meters. $2L + 2W = 100$.

The farmer cannot make the length and width infinitely large. They are constrained by the amount of fence they have.

4.1 The Method of Lagrange Multipliers

This is a powerful technique for solving constrained optimization problems. The core intuition is that at the optimal point, the contour line of the objective function will be **tangent** to the curve of the constraint.



At this point of tangency, the gradient vectors of the objective function (f) and

the constraint function (g) must be parallel. This means one is a scalar multiple of the other:

$$\nabla f(x, y) = \lambda \nabla g(x, y) \tag{1}$$

Here, λ (lambda) is the **Lagrange Multiplier**. This insight allows us to convert a constrained problem into an unconstrained one by solving a system of equations.

Example 1: A Simple Case

Maximize $f(x, y) = xy$ subject to the constraint $x + y = 6$.

1. **Identify functions:** $f(x, y) = xy$ and $g(x, y) = x + y - 6 = 0$.
2. **Find Gradients:** $\nabla f = \langle y, x \rangle$ and $\nabla g = \langle 1, 1 \rangle$.
3. **Set up Lagrange Equations** ($\nabla f = \lambda \nabla g$):

$$y = \lambda(1) \implies y = \lambda$$

$$x = \lambda(1) \implies x = \lambda$$

This tells us that at the optimal point, $x = y$.

4. **Solve using the constraint:** Substitute $y = x$ into the constraint equation: $x + x = 6 \implies 2x = 6 \implies x = 3$. Since $x = y$, we have $y = 3$.

The optimal solution is the point **(3, 3)**, which gives a maximum value of $f(3, 3) = 9$.

Example 2: Finding Closest Point on a Surface

Find the point on the plane $x + 2y + 3z = 6$ that is closest to the origin $(0, 0, 0)$.

1. **Identify functions:**

- We want to minimize the distance to the origin. To make the math easier, we can minimize the *squared distance*, as this will occur at the same point.
- Objective: $f(x, y, z) = d^2 = x^2 + y^2 + z^2$.
- Constraint: $g(x, y, z) = x + 2y + 3z - 6 = 0$.

2. **Find Gradients:**

- $\nabla f = \langle 2x, 2y, 2z \rangle$
- $\nabla g = \langle 1, 2, 3 \rangle$

3. **Set up Lagrange Equations** ($\nabla f = \lambda \nabla g$):

$$2x = \lambda(1) \implies x = \lambda/2$$

$$2y = \lambda(2) \implies y = \lambda$$

$$2z = \lambda(3) \implies z = 3\lambda/2$$

This gives us the relationship between x, y , and z at the optimal point.

4. **Solve using the constraint:** Substitute these expressions for x, y , and z into the constraint equation:

$$(\lambda/2) + 2(\lambda) + 3(3\lambda/2) = 6$$

$$\lambda/2 + 2\lambda + 9\lambda/2 = 6$$

$$\lambda/2 + 4\lambda/2 + 9\lambda/2 = 6$$

$$14\lambda/2 = 6$$

$$7\lambda = 6 \implies \lambda = 6/7$$

5. **Find the optimal point:** Now substitute the value of λ back into our expressions for x, y, z :

- $x = \lambda/2 = (6/7)/2 = 3/7$
- $y = \lambda = 6/7$
- $z = 3\lambda/2 = 3(6/7)/2 = 9/7$

The point on the plane closest to the origin is $(\frac{3}{7}, \frac{6}{7}, \frac{9}{7})$.